

INTERMEDIATE LEVEL LINEAR ALGEBRA NOTES

1. VECTOR CALCULATIONS

Norm (Length of a Vector):

For $v = (v_1, v_2, \dots, v_n)$,

$$\|v\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

Example: If $v = (3, 4)$, $\|v\| = 5$

Angle Between Two Vectors:

Dot Product: $u \cdot v = u_1v_1 + u_2v_2 + \dots + u_nv_n$

$$\cos(\theta) = (u \cdot v) / (\|u\| \|v\|)$$

2. EIGENVALUES OF SPECIAL MATRICES

Diagonal Matrix:

Eigenvalues are the entries on the main diagonal.

Example: $\text{diag}(3, 5, 7) \rightarrow \text{eigenvalues} = 3, 5, 7$

Symmetric Matrix:

If $A = A^T$ (real symmetric), then:

- All eigenvalues are real
- Eigenvectors of distinct eigenvalues are orthogonal

3. DIAGONALIZABILITY

A matrix A is diagonalizable if:

$$A = P D P^{-1}$$

Where D is diagonal matrix of eigenvalues.

A matrix is guaranteed diagonalizable if:

- It has n distinct eigenvalues
- It is real symmetric

4. ORTHOGONAL COMPLEMENT & PROJECTIONS

Orthogonal Complement ($W^\perp$):

Contains all vectors perpendicular to every vector in subspace W .

$$\dim(W) + \dim(W^\perp) = n$$

Projection Formula:

$$\text{proj}_v(u) = \frac{u \cdot v}{v \cdot v} v$$

5. DETERMINANT PROPERTY

The determinant of a matrix equals the product of its eigenvalues.

If eigenvalues are $\lambda_1, \lambda_2, \dots, \lambda_n$:

$$\det(A) = \lambda_1 \times \lambda_2 \times \dots \times \lambda_n$$